WORKED SOLUTION

# STEP II 1994, Question 4

Pure • Areas of Ellipses

**STEP II 1994, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q4

## The key idea

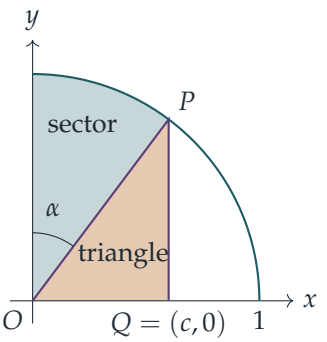
The first integral is an area under a quarter circle, and that area splits cleanly into a right-angled triangle and a circular sector, so no substitution is needed. An ellipse is a circle stretched by different factors in the two directions, and stretching multiplies areas by a constant, which gives the second result at once. For the final part, symmetry does almost all the work: the two ellipses are reflections of each other in the line $y = x$, so we only need one-eighth of the intersection, and that piece is the second result minus a triangle.

## Solution

### The integral of $(1 - x^2)^{1/2}$

The region in question is bounded by the quarter circle $y = (1 - x^2)^{1/2}$, the axes and the line $x = c$. (The curve has to be read as the boundary: the region is the set of points with $0 \leq y \leq (1 - x^2)^{1/2}$ and $0 \leq x \leq c$.) Its area is exactly the integral we want.

Let $P = (c, (1 - c^2)^{1/2})$ be the point of the circle above $x = c$, let $Q = (c, 0)$, and let $\alpha = \sin^{-1} c$, so that $0 < \alpha \leq \frac{\pi}{2}$. Joining $O$ to $P$ cuts the region into two pieces.



- The triangle $OQP$ has a right angle at $Q$, base $c$ and height $(1-c^2)^{1/2}$, so its area is $\frac{1}{2}c(1-c^2)^{1/2}$.
- The rest is the sector of the unit circle between $OP$ and the $y$-axis. The angle between $OP$ and the $y$-axis is $\alpha$, because the horizontal distance of $P$ from the $y$-axis is $c = 1 \cdot \sin\alpha$. A sector of radius 1 and angle $\alpha$ has area $\frac{1}{2}\alpha = \frac{1}{2}\sin^{-1} c$.

Adding the two pieces,

$$\int_0^c (1-x^2)^{\frac{1}{2}}\,\mathrm{d}x = \tfrac{1}{2}\left[c(1-c^2)^{\frac{1}{2}} + \sin^{-1} c\right], \qquad 0 < c \le 1,$$

as required. (When $c = 1$ the triangle is flat and we get the quarter-circle area $\frac{\pi}{4}$.)

#### The area under the ellipse

Again the region lies under the upper half of the curve, $y = b\left(1 - x^2/a^2\right)^{1/2}$, for $0 \le x \le c$, so its area is

$$\int_0^c b\left(1 - \frac{x^2}{a^2}\right)^{\frac{1}{2}}\mathrm{d}x.$$

Substituting $x = au$, so that $\mathrm{d}x = a\,\mathrm{d}u$ and the limits become 0 and $c/a$ (with $0 < c/a \le 1$, since $0 < c \le a$), this is

$$ab\int_0^{c/a} (1-u^2)^{\frac{1}{2}}\,\mathrm{d}u = \frac{ab}{2}\left[\frac{c}{a}\left(1 - \frac{c^2}{a^2}\right)^{\frac{1}{2}} + \sin^{-1}\left(\frac{c}{a}\right)\right],$$

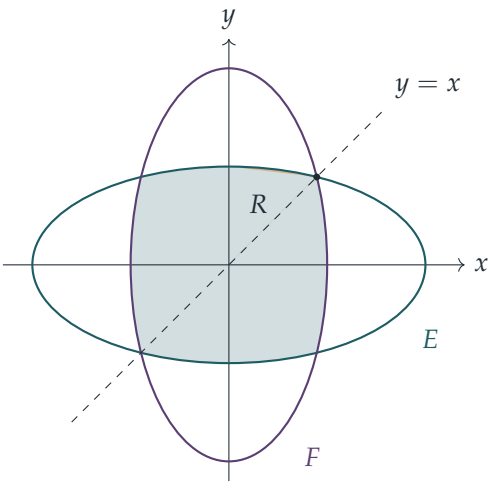
by the first result with $c$ replaced by $c/a$. This is the required area.

Geometrically: the ellipse is the unit circle stretched by $a$ horizontally and $b$ vertically, and that stretch multiplies every area by $ab$.

#### The intersection of the two ellipses

Both $E$ and $F$ are unchanged by $x \mapsto -x$ and by $y \mapsto -y$, so $E \cap F$ is too, and its area is four times the area of its part in the quadrant $x, y \ge 0$. Also, swapping $x$ and $y$ turns $E$ into $F$ and $F$ into $E$, so $E \cap F$ is symmetric in the line $y = x$. So the quarter in the first quadrant is made of two mirror-image halves, and

$$\text{area}(E \cap F) = 8 \times \text{area of } R, \qquad R = \{(x,y) \in E \cap F :\ 0 \le x \le y\}.$$




**Ask yourself** In $R$, which of the two ellipses actually forms the boundary?

For any point,

$$\left(\frac{x^2}{b^2}+\frac{y^2}{a^2}\right)-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)=(x^2-y^2)\left(\frac{1}{b^2}-\frac{1}{a^2}\right).$$

When $0 \le x \le y$ we have $x^2 - y^2 \le 0$, and since $0 < b \le a$ we have $\frac{1}{b^2} - \frac{1}{a^2} \ge 0$. So the difference is $\le 0$: on that side of $y = x$, any point in $E$ is automatically in $F$. Hence

$$R = \{(x, y) \in E : \ 0 \le x \le y\},$$

and only $E$ matters. (This is where $b \le a$ is used; if $b > a$ the roles of the ellipses swap.)

The boundary of $E$ meets $y = x$ in the first quadrant where $x^2\left(\frac{1}{a^2} + \frac{1}{b^2}\right) = 1$, that is at $x = y = x_0$ with

$$x_0 = \frac{ab}{\sqrt{a^2+b^2}}.$$

For $0 \le x \le x_0$ the upper half of $E$ lies on or above $y = x$ (it starts at height $b > 0$, and meets the line only at $x_0$), while for $x > x_0$ it lies below the line. So $R$ is the region between $y = x$ and the curve $y = b(1 - x^2/a^2)^{1/2}$ for $0 \le x \le x_0$:

$$\text{area}(R) = \underbrace{\int_0^{x_0} b\left(1-\frac{x^2}{a^2}\right)^{\frac{1}{2}}\mathrm{d}x}_{\text{previous result with } c=x_0} - \underbrace{\tfrac{1}{2}x_0^2}_{\text{triangle under } y=x}.$$

With $c = x_0$ (and $0 < x_0 < a$, as needed),

$$\frac{c}{a}=\frac{b}{\sqrt{a^2+b^2}}, \qquad \left(1-\frac{c^2}{a^2}\right)^{\frac{1}{2}}=\frac{a}{\sqrt{a^2+b^2}}, \qquad \frac{c}{a}\left(1-\frac{c^2}{a^2}\right)^{\frac{1}{2}}=\frac{ab}{a^2+b^2}.$$

So

$$\begin{aligned}\text{area}(E \cap F) &= 8\,\text{area}(R)\\ &= 4ab\left[\frac{ab}{a^2+b^2} + \sin^{-1}\Big(\frac{b}{\sqrt{a^2+b^2}}\Big)\right] - \frac{4a^2b^2}{a^2+b^2}\\ &= 4ab\sin^{-1}\Big(\frac{b}{\sqrt{a^2+b^2}}\Big),\end{aligned}$$

since the two algebraic terms cancel exactly. This is the required area.

**Ask yourself** Why do the algebraic terms cancel?

The term $\frac{ab}{2}\cdot\frac{c}{a}(1-c^2/a^2)^{1/2} = \frac{1}{2}x_0y_0$ is the triangle $O$, $(x_0,0)$, $(x_0,y_0)$ from the circle picture, stretched. Here $y_0 = x_0$, so it is exactly the triangle under $y = x$ that we subtract. What is left is a stretched sector, which is why the answer is a pure inverse sine.

## Checks

- If $a = b$ the ellipses are the same circle, and the formula gives $4a^2\sin^{-1}\frac{1}{\sqrt{2}} = 4a^2\cdot\frac{\pi}{4} = \pi a^2$, as it should.
- As $b \to 0$ with $a$ fixed, the area behaves like $4ab\cdot\frac{b}{a} = 4b^2$: the intersection is nearly the square of side $2b$ where $E$'s thin horizontal strip crosses $F$'s thin vertical one.
- For $a = 2$, $b = 1$ the formula gives $8\sin^{-1}\frac{1}{\sqrt{5}} \approx 3.70918$. Simpson's rule on the lower of the two boundaries gives 3.70918, and a Monte Carlo estimate with $2\times10^5$ points gives 3.707. The first two results were also checked by quadrature for many random $a$, $b$, $c$.

## Common slips

- Getting the sector angle wrong: the sector is between $OP$ and the $y$-axis, with angle $\sin^{-1}c$, not $\cos^{-1}c$.
- Not checking which ellipse bounds the region. In the half $0 \le x \le y$ it is $E$ only because $b \le a$; with the other choice the answer would come out with $a$ in place of $b$ inside the inverse sine.

## Beyond STEP

A linear map $(x,y)\mapsto(ax,by)$ multiplies every area by $ab$, and in general a linear map multiplies areas by the absolute value of its determinant. This is the two-dimensional case of the Jacobian factor in the change-of-variables formula for multiple integrals,

which you meet in first-year calculus. Strang's *Introduction to Linear Algebra* explains the determinant as an area (and volume) scale factor.

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